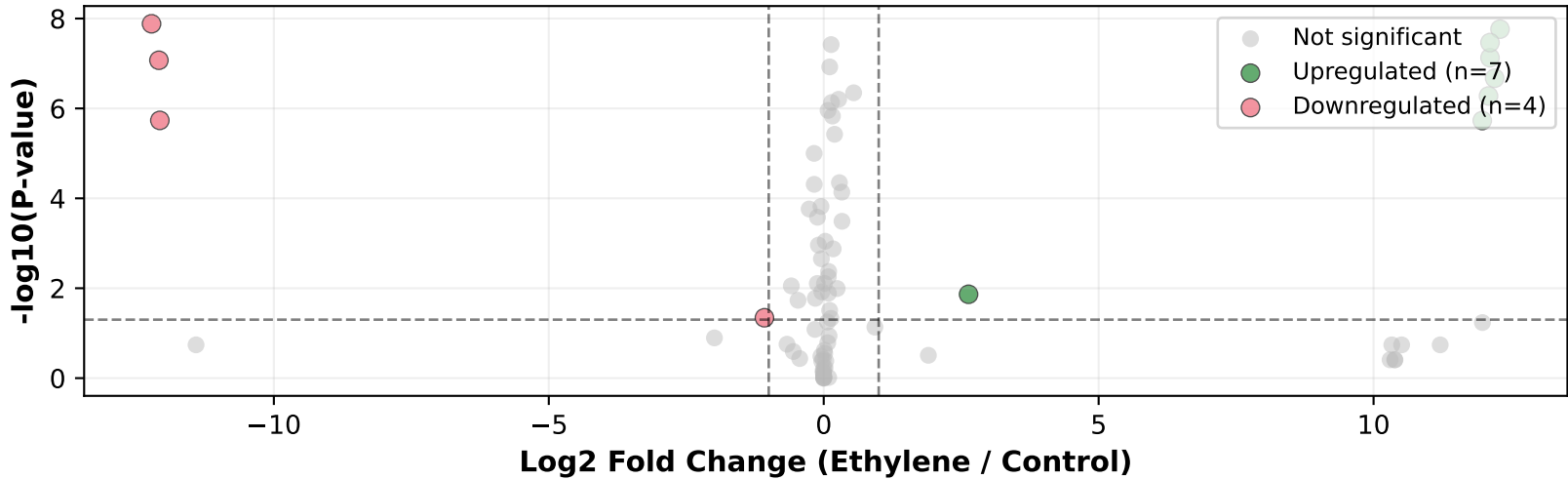
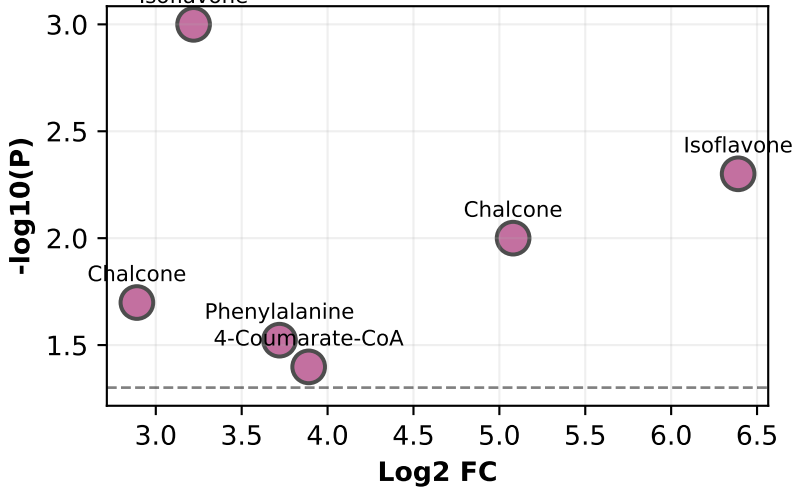


Multi-Omics Integration: Ethylene-Induced Metabolic Reprogramming

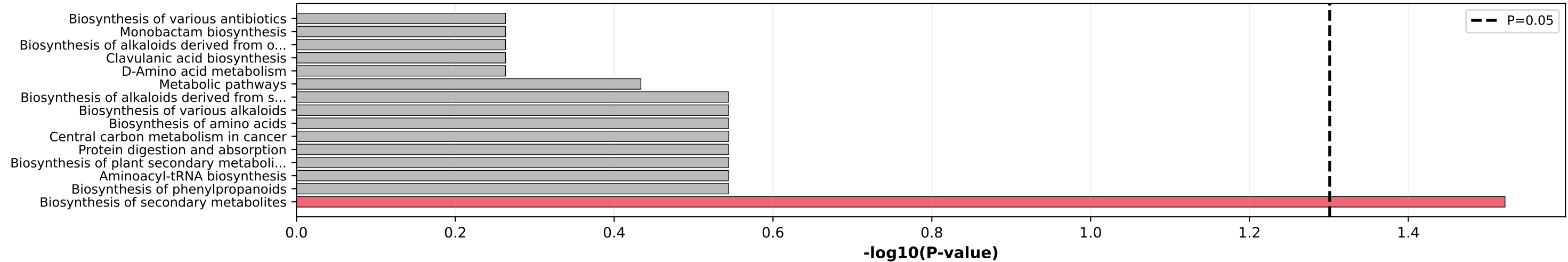
A. Metabolomics: Volcano Plot



B. Proteomics



C. KEGG Pathway Enrichment



MULTI-OMICS INTEGRATION SUMMARY

Metabolomics: 7 metabolites upregulated, 4 downregulated ($P < 0.05$, $|\text{Log2FC}| > 1$)
Top hits: Daidzein (12.3 $\times$), Formononetin (12.2 $\times$), Malonylgenistin (12.1 $\times$)

Proteomics: 6 key enzymes upregulated in isoflavonoid pathway
IFR (6.4 $\times$), CHI (5.1 $\times$), 4CL (3.9 $\times$), PAL (3.7 $\times$), IFS (3.2 $\times$), CHS (2.9 $\times$)

Pathway: KEGG map01110 (Biosynthesis of secondary metabolites) $P = 0.030$ ***
PlantCyc ISOFLAVONOID-SYN and SECONDARY-METABOLITE-BIOSYNTHESIS concordant

Conclusion: Ethylene induces coordinated activation of isoflavonoid biosynthesis at both the metabolite AND protein levels, representing a defense response.